COSC363-25S1: Computer Graphics

Assignment 2 - Ray Tracer Report

Done by: Yanlong Su

Student ID: 57817379

Declaration

I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work.

Name: Yanlong Su

Student ID: 57817379

Date: 24.May.2025

Contents

1.0 Introduction 2

2.0 Basic ray tracer 2

3.0 Additional features 2

3.1 Anti-aliasing 2

3.2 Texture on sphere object 2

3.3 Other features 2

4.0 Compiling and running the program 4

5.0 Reference 4

6.0 Generative AI 4

1.0 Introduction

The rendered scene is produced by a ray tracing program that is capable of handling various geometric objects and global illumination effects. Advanced features, such as stochastic sampling and constructive solid geometry, are not supported.

On the laboratory machines, it takes approximately 25 seconds to generate the output if adaptive sampling is turned on. The image resolution is 500×500 pixels, with a ray tracing recursion depth of 25 and an anti-aliasing depth of 1 in adaptive sampling.

2.0 Basic ray tracer

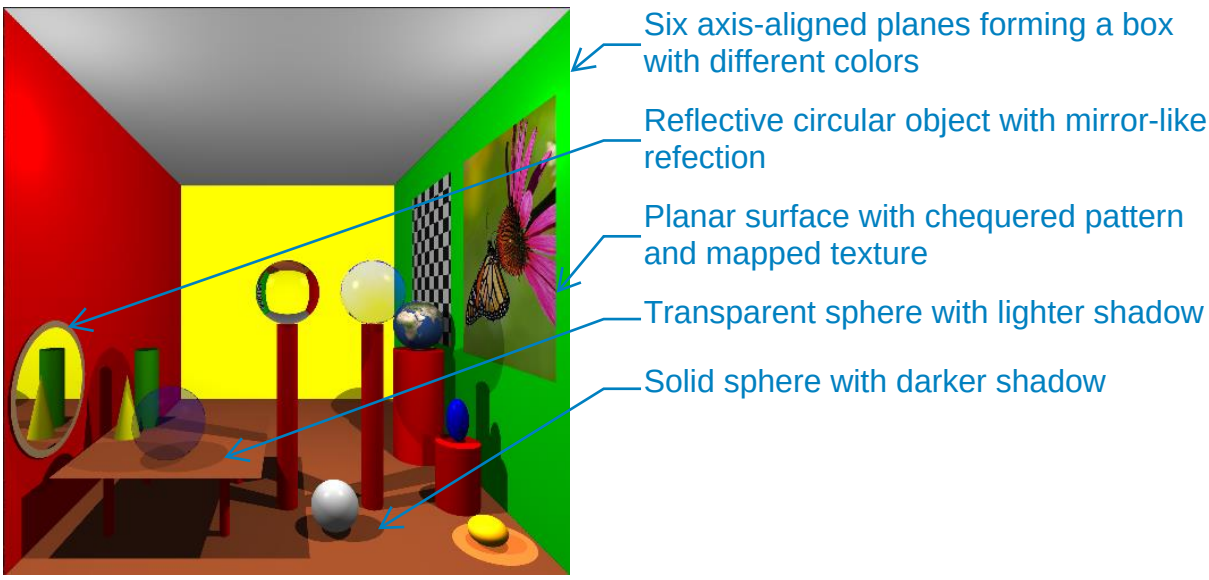


Figure 1 Basic ray tracer features implemented

3.0 Additional features

3.1 Anti-aliasing

This ray tracer can switch between simple sampling and adaptive sampling (line 367-368 in RayTracer.cpp). When adaptive sampling is used, a cell is subdivided only when the color of a ray differs significantly from that of its neighboring rays. Refer below Figure 3.

3.2 Texture on sphere object

This ray tracer uses UV mapping. It first determines the point where the ray intersects the sphere, then converts that point into UV coordinates. Finally, it uses the (u, v) values to retrieve the corresponding color from the texture. Note that both u and v are within range [0, 1]. Refer below Figure 2.

$$u = 0.5 + \frac{arctan2(dz,dx)}{2\pi}, \quad v = 0.5 + \frac{arcsin(dy)}{\pi}$$

3.3 Other features

All other additional features exactly follow the methods / algorithms given in the lecture notes.

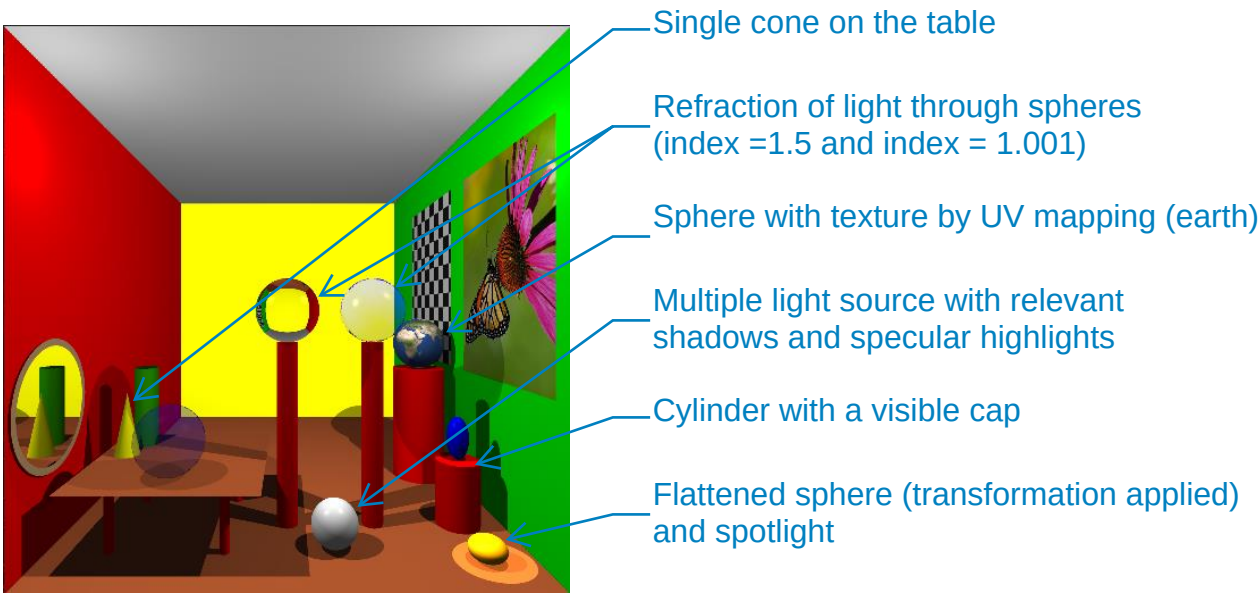
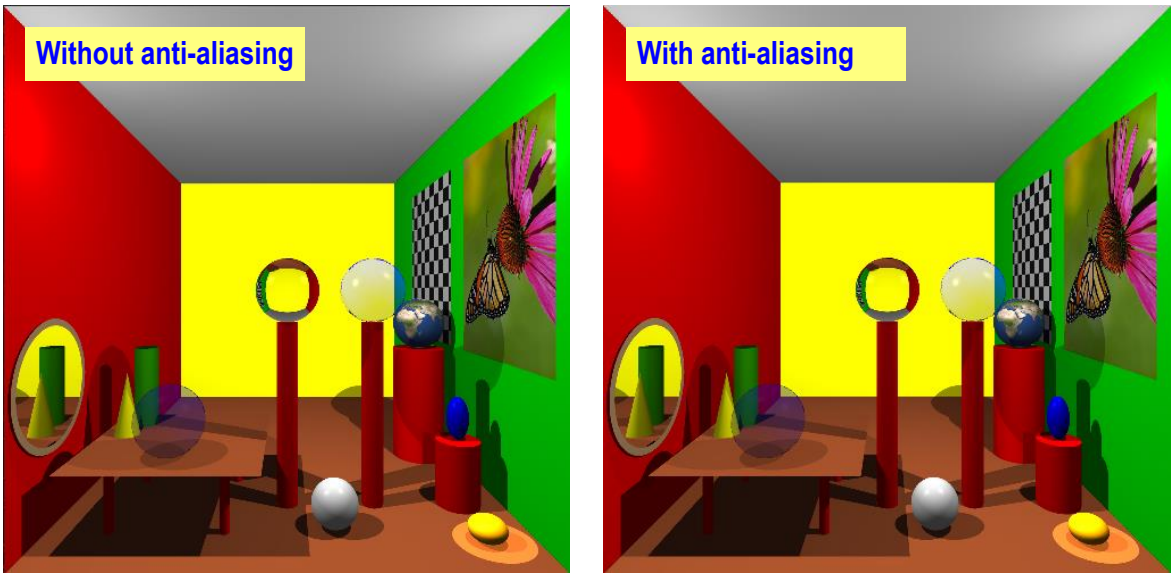


Figure 2 Extra ray tracer features implemented



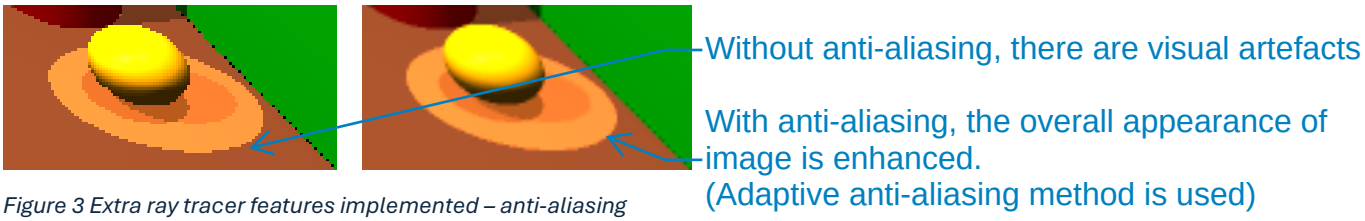


Figure 3 Extra ray tracer features implemented – anti-aliasing

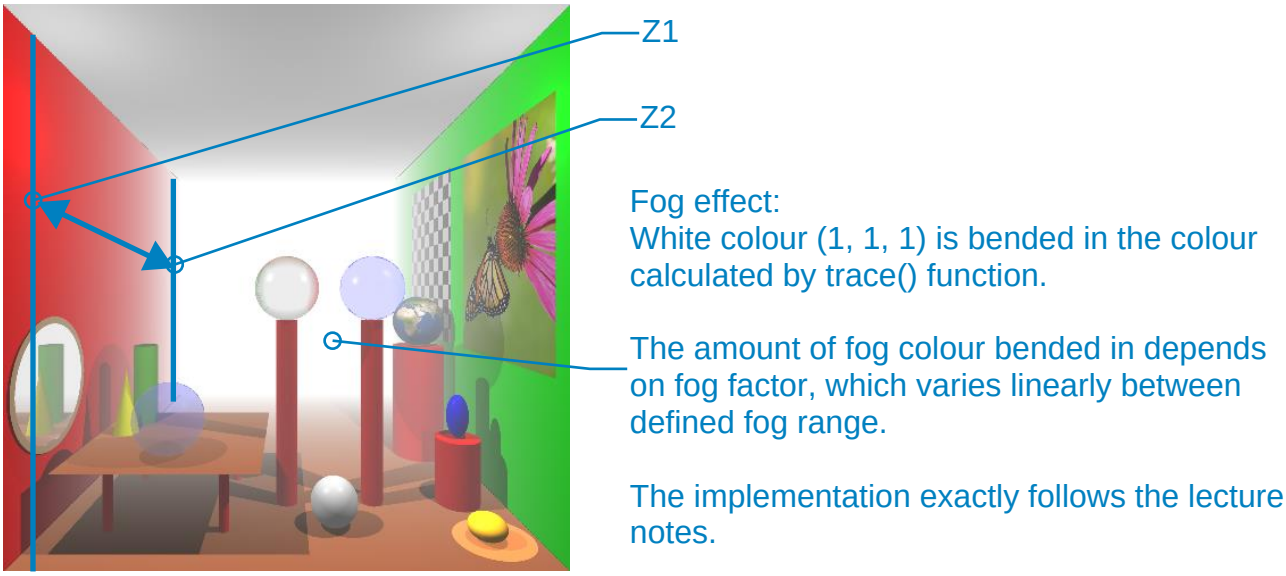


Figure 4 Extra ray tracer features implemented – fog

4.0 Compiling and running the program

The best way to compile and run the program is by using CMake and CMake Tools. Place all source files and the provided CMakeLists.txt in a folder. Then, either use the CMake UI panel or run the following command in a terminal: *make && ./ RayTracer*

5.0 Reference

Textures are either from the COSC363 lab materials or downloaded from Polyhaven.

All algorithms and design patterns follow the labs of COSC363.

6.0 Generative AI

ChatGPT is used to answer questions about debugging, explain the usage of computer graphics functions, and perform grammar checking for both codes and reports.